

# Google Advanced Data Analytics Capstone Project: Develop a model to predict employee departure

## Problem statement

High rate of turnover among Salifort Motors (multinational vehicle manufacturing corporation) employees, which is costly in the financial sense.

- What we know - employee survey results from HR, see Data dictionary
- What we do not know - what's driving the turnover

## Stakeholders

Senior management and HR

## Goal

To tackle the problem, stakeholders want to learn about what might be driving the turnover and predict whether an employee will leave the company – to do this, the team aims to develop a *classification* model to predict employee departure.

## Project milestones

1. [EDA & data visualisations](#) (includes statistics, data cleaning, ethical considerations)
2. [Logistic regression modeling](#)
3. [Tree-based modeling](#) - Random forest and XGBoost
4. [Summary & recommendations](#)

## Data dictionary

Dataset: *HR\_capstone\_dataset.csv*

- 14,999 rows – each row is a different employee's self-reported information
- 10 columns of self-reported information from employees

| Column name           | Type  | Description                                                       |
|-----------------------|-------|-------------------------------------------------------------------|
| satisfaction_level    | int64 | The employee's self-reported satisfaction level [0-1]             |
| last_evaluation       | int64 | Score of employee's last performance review [0-1]                 |
| number_project        | int64 | Number of projects employee contributes to                        |
| average_monthly_hours | int64 | Average number of hours employee worked per month                 |
| time_spend_company    | int64 | How long the employee has been with the company (years)           |
| work_accident         | int64 | Whether or not the employee experienced an accident while at work |
| left                  | int64 | Whether or not the employee left the company                      |
| promotion_last_5years | int64 | Whether or not the employee was promoted in the last 5 years      |
| department            | str   | The employee's department                                         |
| salary                | str   | The employee's salary (low, medium, or high)                      |

# EDA & data visualisations

## Ethical considerations

- Data has been desensitised - does not reveal employee identity or salary
- Data is company-owned
- Office locations of employees are unknown - results may not be representative of every location
- Post-cleaning data contains 11k observations - adequate sample size

## Target variable, y (categorical)

- People who left: 'left' = 1
- People who stayed: 'left' = 0
- Left: 1991 persons, 16.6% of observations
- Stayed: 10000 persons, 83.4% of observations
- Class imbalance but not too extreme

## Evaluation metrics

- False positives: Model predicts employee is leaving when it is not true  
→ incur retention costs and does not offer high value outcome
- False negatives: Model predicts employee is staying when it is not true  
→ employee departure due to retention action not taken in time
- To minimise FP and FN counts
- Look into F1 score - harmonic mean of precision and recall, which takes into account both FP FN

### Metrics:

- **Precision** measures the proportion of data points predicted as True that are actually True. The proportion of positive predictions that are true positives.
- **Recall** measures the proportion of data points that are predicted as True, out of all the data points that are actually True. The proportion of positives that are correctly classified.
- **F1-score** is an aggregation of precision and recall.
- **Accuracy** measures the proportion of data points that are correctly classified.
- **AUC** is the area under the ROC (Receiver Operating Characteristic) curve; it's also considered the probability that the model ranks a random positive example more highly than a random negative example.

## Descriptive statistics

|               | satisfacti<br>on_level | last_evalu<br>ation | number_p<br>roject | average_<br>montly_h<br>ours | time_spe<br>nd_comp<br>any | Work_acc<br>ident | left             | promotio<br>n_last_5y<br>ears | Departme<br>nt | salary |
|---------------|------------------------|---------------------|--------------------|------------------------------|----------------------------|-------------------|------------------|-------------------------------|----------------|--------|
| <b>count</b>  | 14999.000<br>000       | 14999.000<br>000    | 14999.000<br>000   | 14999.000<br>000             | 14999.000<br>000           | 14999.000<br>000  | 14999.000<br>000 | 14999.000<br>000              | 14999          | 14999  |
| <b>unique</b> | NaN                    | NaN                 | NaN                | NaN                          | NaN                        | NaN               | NaN              | NaN                           | 10             | 3      |
| <b>top</b>    | NaN                    | NaN                 | NaN                | NaN                          | NaN                        | NaN               | NaN              | NaN                           | sales          | low    |
| <b>freq</b>   | NaN                    | NaN                 | NaN                | NaN                          | NaN                        | NaN               | NaN              | NaN                           | 4140           | 7316   |
| <b>mean</b>   | 0.612834               | 0.716102            | 3.803054           | 201.05033<br>7               | 3.498233                   | 0.144610          | 0.238083         | 0.021268                      | NaN            | NaN    |
| <b>std</b>    | 0.248631               | 0.171169            | 1.232592           | 49.943099                    | 1.460136                   | 0.351719          | 0.425924         | 0.144281                      | NaN            | NaN    |
| <b>min</b>    | 0.090000               | 0.360000            | 2.000000           | 96.000000                    | 2.000000                   | 0.000000          | 0.000000         | 0.000000                      | NaN            | NaN    |
| <b>25%</b>    | 0.440000               | 0.560000            | 3.000000           | 156.00000<br>0               | 3.000000                   | 0.000000          | 0.000000         | 0.000000                      | NaN            | NaN    |
| <b>50%</b>    | 0.640000               | 0.720000            | 4.000000           | 200.00000<br>0               | 3.000000                   | 0.000000          | 0.000000         | 0.000000                      | NaN            | NaN    |
| <b>75%</b>    | 0.820000               | 0.870000            | 5.000000           | 245.00000<br>0               | 4.000000                   | 0.000000          | 0.000000         | 0.000000                      | NaN            | NaN    |
| <b>max</b>    | 1.000000               | 1.000000            | 7.000000           | 310.00000<br>0               | 10.000000                  | 1.000000          | 1.000000         | 1.000000                      | NaN            | NaN    |

## Unique values

- Work\_accident: [0 1]
- left: [1 0]
- promotion\_last\_5years: [0 1]
- Department: ['sales' 'accounting' 'hr' 'technical' 'support' 'management' 'IT' 'product\_mng' 'marketing' 'RandD']
- salary: ['low' 'medium' 'high']

## Data cleaning

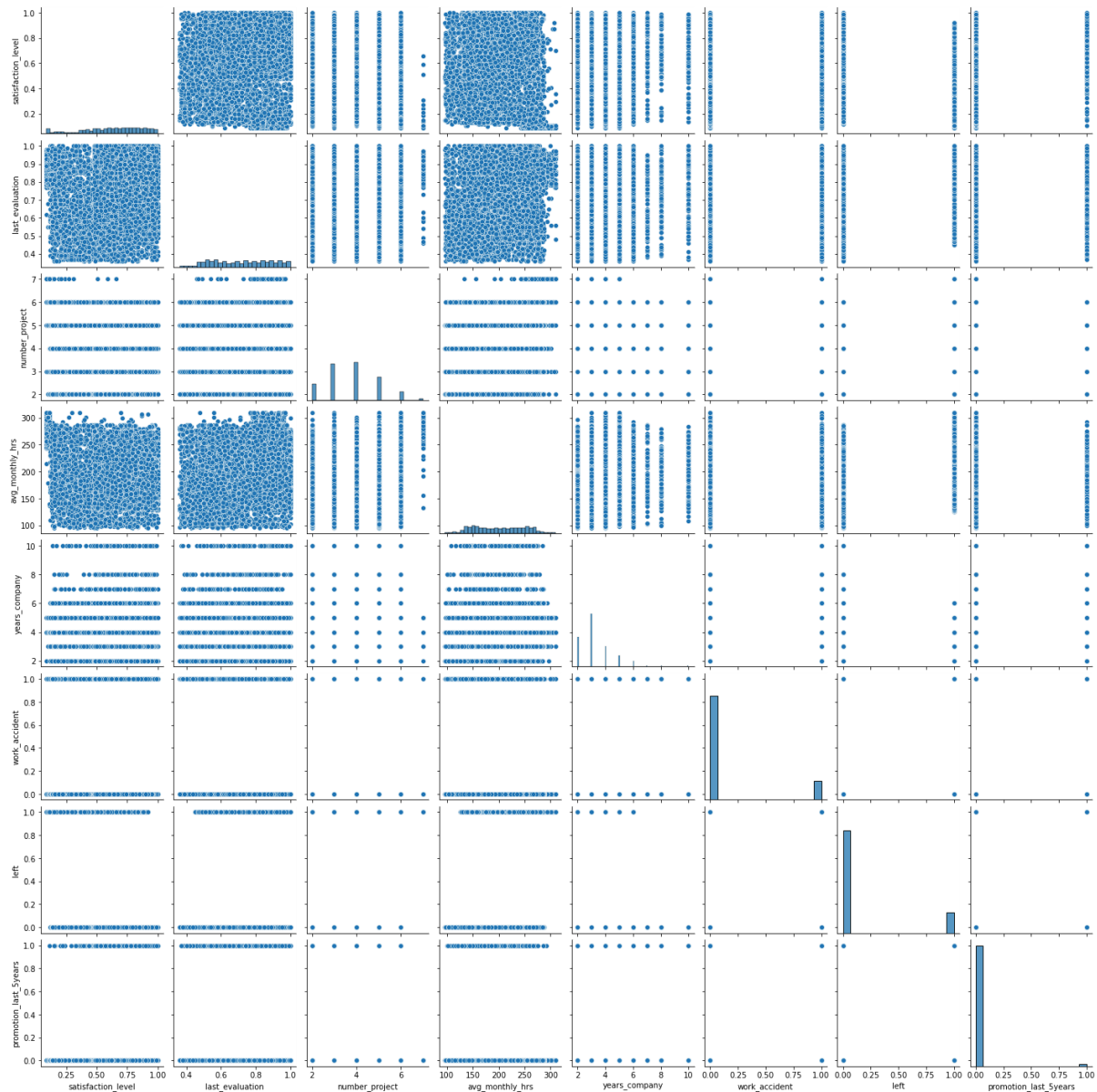
- Renamed columns
  - average\_montly\_hours → avg\_monthly\_hrs
  - time\_spend\_company → years\_company
  - Work\_accident → work\_accident
  - Department → dept
  - Resulting df1 columns:  
['satisfaction\_level', 'last\_evaluation', 'number\_project', 'avg\_monthly\_hrs', 'years\_company', 'work\_accident', 'left', 'promotion\_last\_5years', 'dept', 'salary']
- Dropped rows that are exact duplicates (chances of exact duplicates being different employees are very low)
  - Dropped 3008 rows
  - Resulting df1 shape (11991, 10)

## Outliers

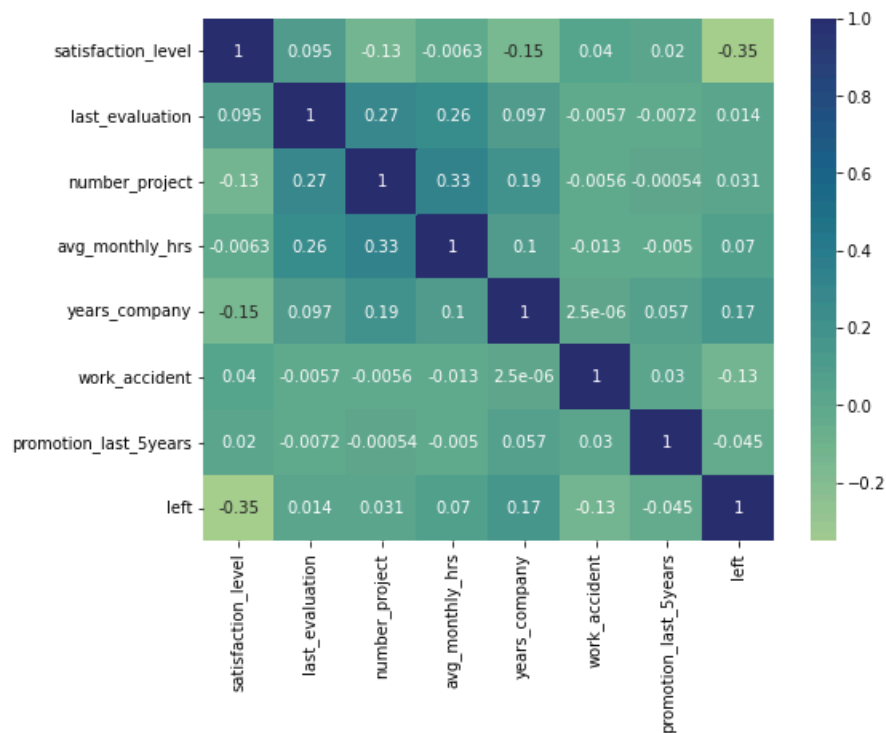
- Must be removed for logistic regression
- `df1['years_company'] > upper_limit`, where `upper_limit = 5.5`
- 824 observations

## Data visualisations

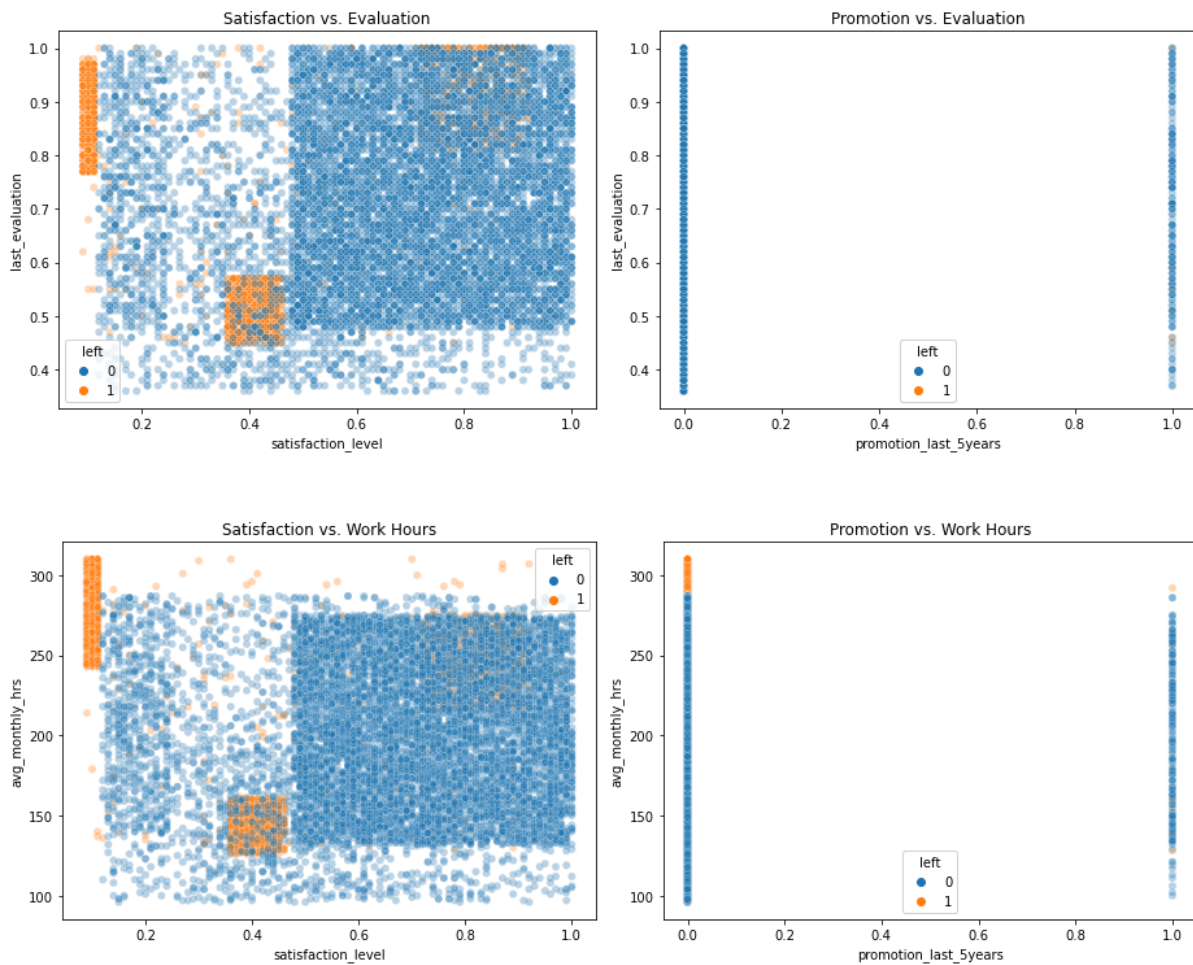
- Pairplots: examine relationships between variables  
→ multicollinearity not observed



- Heatmap: values nearer to +/-1.0 indicate +/- strong correlation between variables  
 → people who left is negatively correlated with their satisfaction level  
 → similarly for no. of years in the company and no. of projects

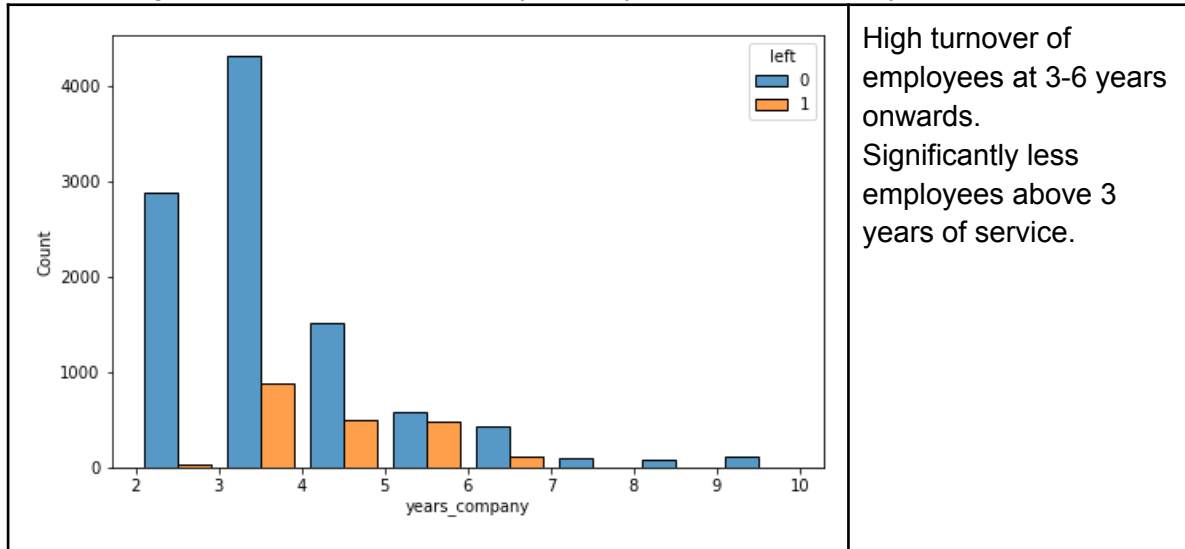


- Scatterplots: examine areas of concentration or non-overlapping

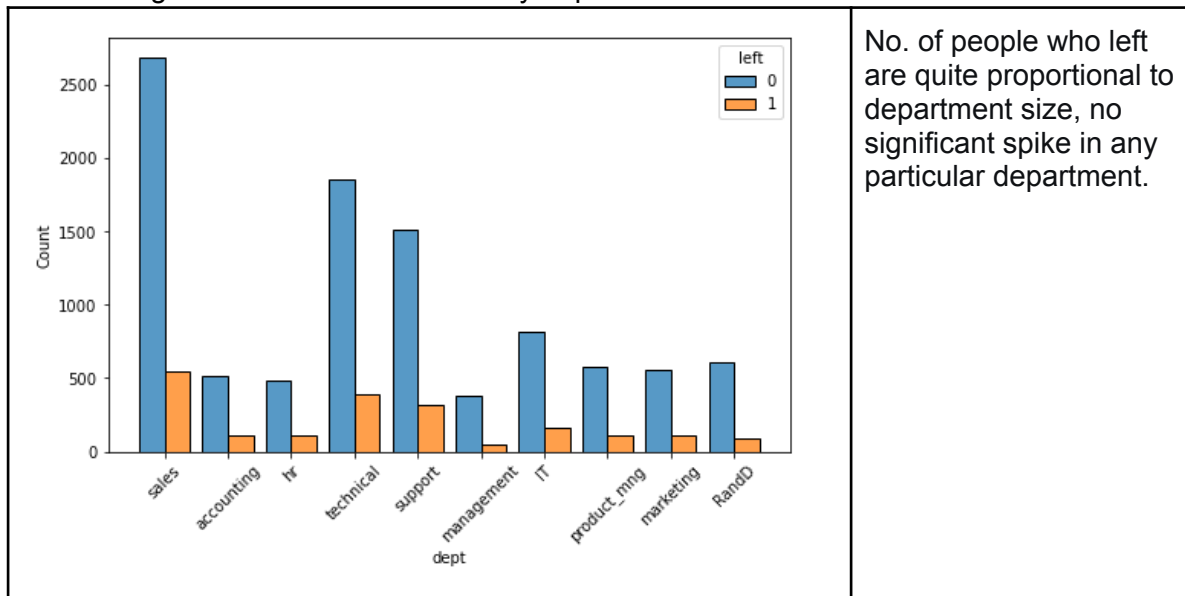


| left | promotion_last_5years | avg_monthly_hrs |        | satisfaction_level |        |                                                                                     |
|------|-----------------------|-----------------|--------|--------------------|--------|-------------------------------------------------------------------------------------|
|      |                       | mean            | median | mean               | median |                                                                                     |
| 0    | 0                     | 198.932585      | 198.0  | 0.667250           | 0.690  | People who were not promoted and left:<br>Highest work hours<br>Lowest satisfaction |
|      | 1                     | 199.451282      | 203.0  | 0.673128           | 0.700  |                                                                                     |
| 1    | 0                     | 208.281392      | 226.0  | 0.440061           | 0.410  |                                                                                     |
|      | 1                     | 178.625000      | 157.0  | 0.492500           | 0.445  |                                                                                     |

- Histogram: examine distribution by no. of years in the company



- Histogram: examine distribution by departments



# Logistic regression modeling

## Assumptions

- Outcome variable is categorical → 'left' is categorical with unique values [1, 0]
- Observations are independent of each other → each row represents a unique employee
- No severe multicollinearity among X variables → as shown through pariplots
- No extreme outliers → outliers to be removed
- Linear relationship between each X variable and the logit of the outcome variable → to check *coef* after fitting model to data
- Sufficiently large sample size → 11k observations

## Data prep

- No null for regression modeling - df has no missing value
- Filter out outliers as identified during EDA
- Encode 'dept' → 10 dept columns [0,1]
- Encode 'salary' → map({"low": 1, "medium": 2, "high": 3})
- Outcome variable y = 'left'

## Results & evaluation

X.columns:

```
['satisfaction_level', 'last_evaluation', 'number_project', 'avg_monthly_hrs',  
'years_company', 'work_accident', 'promotion_last_5years', 'salary',  
'dept_IT', 'dept_RandD', 'dept_accounting', 'dept_hr',  
'dept_management', 'dept_marketing', 'dept_product_mng', 'dept_sales',  
'dept_support', 'dept_technical']
```

coef:

```
[[-4.35872176e+00 -8.83710770e-02 -4.36629694e-01 3.24585095e-03  
1.02583699e+00 -1.45921646e+00 -4.85278058e-01 -5.07214405e-01  
2.68951379e-02 -3.07823645e-01 -1.70901174e-01 5.30719479e-02  
-1.95137118e-01 -3.98193459e-02 -1.12096882e-01 6.86662993e-02  
-3.78872499e-02 5.23445290e-02]]
```

intercept:

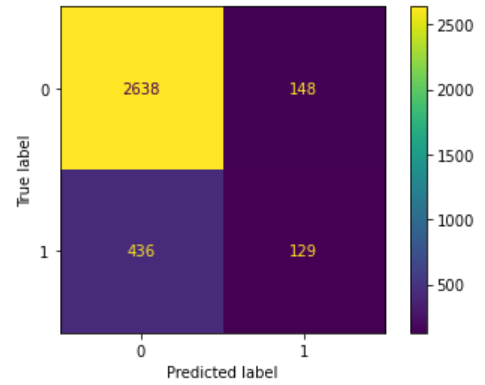
```
[-0.6626875]
```



### Evaluation metrics

- Accuracy: 0.825724
- Precision: 0.465704
- Recall: 0.228319
- F1: 0.306413
- AUC: 0.587598

|               | precision | recall | f1-score | support |
|---------------|-----------|--------|----------|---------|
| Predict-stay  | 0.86      | 0.95   | 0.90     | 2786    |
| Predict-leave | 0.47      | 0.23   | 0.31     | 565     |
| accuracy      |           |        | 0.83     | 3351    |
| macro avg     | 0.66      | 0.59   | 0.60     | 3351    |
| weighted avg  | 0.79      | 0.83   | 0.80     | 3351    |



- Evaluation metric scores are low, especially precision, recall and f1 → FP and FN counts are high.
- AUC 0.58 indicates the model is only slightly better than a random guess.
- Logistic regression model - poor performance

# Tree-based modeling

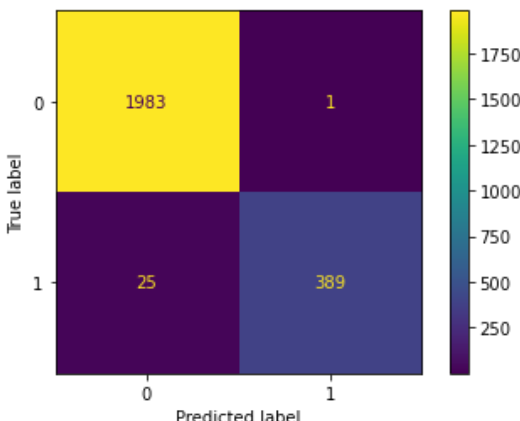
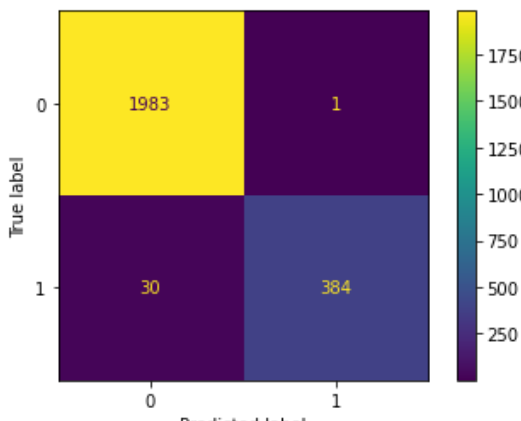
## Assumptions

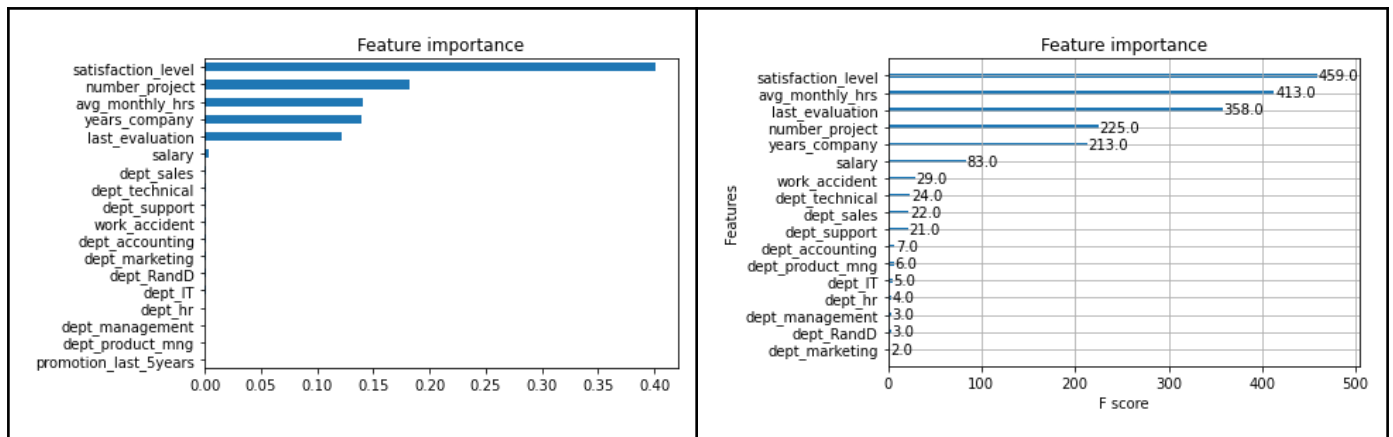
- Nil

## Data prep

- No null for random forest - df has no missing value
- Resilient to outliers → no need to filter out outliers
- Encode 'dept' → 10 dept columns [0,1]
- Encode 'salary' → map({"low": 1, "medium": 2, "high": 3})
- Outcome variable y: 'left'

## Results & evaluation

| Random forest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | XGBoost                                                                                                                                                                                                                                                                                            |           |          |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|----------|---------|--------------|------|------|------|------|---------------|------|------|------|-----|----------|--|--|------|------|-----------|------|------|------|------|--------------|------|------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------|--------|----------|---------|--------------|------|------|------|------|---------------|------|------|------|-----|----------|--|--|------|------|-----------|------|------|------|------|--------------|------|------|------|------|
| <ul style="list-style-type: none"><li>• GridSearchCV refit='f1'</li><li>• Best score: 0.9546599496221662</li><li>• Best parameters:<br/>{'max_depth': None,<br/>'max_features': 10,<br/>'min_samples_leaf': 2,<br/>'min_samples_split': 2,<br/>'n_estimators': 100}</li></ul>                                                                                                                                                                                                                                  | <ul style="list-style-type: none"><li>• GridSearchCV refit='f1'</li><li>• Best score: 0.951188986232791</li><li>• Best parameters:<br/>{'colsample_bytree': 0.7,<br/>'learning_rate': 0.1,<br/>'max_depth': 6,<br/>'min_child_weight': 2,<br/>'n_estimators': 100,<br/>'subsample': 0.7}</li></ul> |           |          |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
| <table><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr><tr><td>Predict-stay</td><td>0.99</td><td>1.00</td><td>0.99</td><td>1984</td></tr><tr><td>Predict-leave</td><td>1.00</td><td>0.94</td><td>0.97</td><td>414</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.99</td><td>2398</td></tr><tr><td>macro avg</td><td>0.99</td><td>0.97</td><td>0.98</td><td>2398</td></tr><tr><td>weighted avg</td><td>0.99</td><td>0.99</td><td>0.99</td><td>2398</td></tr></table> |                                                                                                                                                                                                                                                                                                    | precision | recall   | f1-score | support | Predict-stay | 0.99 | 1.00 | 0.99 | 1984 | Predict-leave | 1.00 | 0.94 | 0.97 | 414 | accuracy |  |  | 0.99 | 2398 | macro avg | 0.99 | 0.97 | 0.98 | 2398 | weighted avg | 0.99 | 0.99 | 0.99 | 2398 | <table><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr><tr><td>Predict-stay</td><td>0.99</td><td>1.00</td><td>0.99</td><td>1984</td></tr><tr><td>Predict-leave</td><td>1.00</td><td>0.93</td><td>0.96</td><td>414</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.99</td><td>2398</td></tr><tr><td>macro avg</td><td>0.99</td><td>0.96</td><td>0.98</td><td>2398</td></tr><tr><td>weighted avg</td><td>0.99</td><td>0.99</td><td>0.99</td><td>2398</td></tr></table> |  | precision | recall | f1-score | support | Predict-stay | 0.99 | 1.00 | 0.99 | 1984 | Predict-leave | 1.00 | 0.93 | 0.96 | 414 | accuracy |  |  | 0.99 | 2398 | macro avg | 0.99 | 0.96 | 0.98 | 2398 | weighted avg | 0.99 | 0.99 | 0.99 | 2398 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | precision                                                                                                                                                                                                                                                                                          | recall    | f1-score | support  |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
| Predict-stay                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0.99                                                                                                                                                                                                                                                                                               | 1.00      | 0.99     | 1984     |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
| Predict-leave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.00                                                                                                                                                                                                                                                                                               | 0.94      | 0.97     | 414      |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
| accuracy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                    |           | 0.99     | 2398     |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
| macro avg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0.99                                                                                                                                                                                                                                                                                               | 0.97      | 0.98     | 2398     |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
| weighted avg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0.99                                                                                                                                                                                                                                                                                               | 0.99      | 0.99     | 2398     |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | precision                                                                                                                                                                                                                                                                                          | recall    | f1-score | support  |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
| Predict-stay                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0.99                                                                                                                                                                                                                                                                                               | 1.00      | 0.99     | 1984     |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
| Predict-leave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.00                                                                                                                                                                                                                                                                                               | 0.93      | 0.96     | 414      |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
| accuracy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                    |           | 0.99     | 2398     |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
| macro avg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0.99                                                                                                                                                                                                                                                                                               | 0.96      | 0.98     | 2398     |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
| weighted avg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0.99                                                                                                                                                                                                                                                                                               | 0.99      | 0.99     | 2398     |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |
|                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                |           |          |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |           |        |          |         |              |      |      |      |      |               |      |      |      |     |          |  |  |      |      |           |      |      |      |      |              |      |      |      |      |



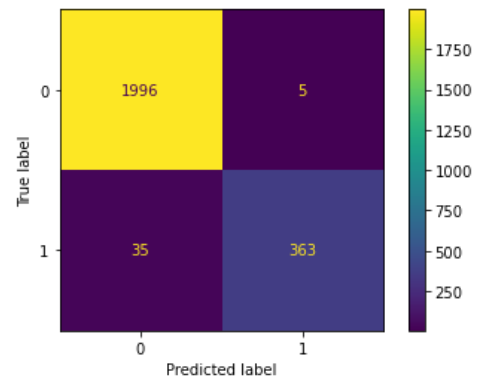
- Both RF and XGB models perform well
- RF performs slightly better than XGB.
- The top 6 important features are the same.

# Summary & recommendations

Champion model: Random forest model

Results of champion model on test data:

|               | precision | recall | f1-score | support |
|---------------|-----------|--------|----------|---------|
| Predict-stay  | 0.98      | 1.00   | 0.99     | 2001    |
| Predict-leave | 0.99      | 0.91   | 0.95     | 398     |
| accuracy      |           |        | 0.98     | 2399    |
| macro avg     | 0.98      | 0.95   | 0.97     | 2399    |
| weighted avg  | 0.98      | 0.98   | 0.98     | 2399    |



- 5 false positives: Model predicts employee is leaving when it is not true  
→ incur retention costs and does not offer high value outcome
- 35 false negatives: Model predicts employee is staying when it is not true  
→ employee departure due to no retention action, incur hiring & retraining costs

## Benefits & limitations

- The model is fairly good in prediction, as shown by the scores above.
- The model requires less computational time/cost as compared to the XGB model.
- However, there could be a handful of employees who are not predicted to leave when in fact they are leaving (false negatives).

## Recommendations

1. Exercise retention actions on good performers predicted leaving, and those groups highlighted in 5a and 5b.
2. Invest time in exploring the top 5 important features i.e. factors that contribute to leaving ['satisfaction\_level', 'last\_evaluation', 'number\_project', 'avg\_monthly\_hrs', 'years\_company']
3. Consider improving the model performance with additional survey data such as the rating of immediate supervisor and branch location.
4. The model uses survey data. The survey is typically conducted annually, and some employees may not complete the survey. Ideally, retention action need not wait til the release of survey results.
5. [Data visualisations](#) reveal higher turnover among
  - a. those with 3-6 years in service → focus on understanding this group of people, as well as those who stayed very long with the company.
  - b. those who worked long hours (>200 hours on average monthly) and were not promoted in recent years → investigate this group of people, whether it is an efficiency problem, over-delegation of work by supervisors or the employee is not rewarded for good performance.